COURSE GUIDE & SYLLABUS · GRADES 7–11 · AGES 12–16

Python (Gr 6–8) Intermediate: Python Game Coders · Level 2

Functions, turtle art, dictionaries and classes: build bigger games in real Python.

The second course in real, typed Python for ages 12–16, picking up where Level 1 ended. In eight live classes learners write functions that return answers, draw with turtle graphics, organise data in dictionaries, lists of records and simple classes, and finish with their own turn-based battle game. Every class ends with a program they can play and show at home: a cricket score calculator, a Diwali rangoli generator, a Ludo dice duel, a Turtle Art Studio, a class quiz bank, Hangman, a virtual pet and a Battle Arena text RPG. Each lesson has a short video that shows the code and its output, a build-along, auto-checked Python Lab challenges, a predict-the-output puzzle, a debugging challenge, a worksheet and a full teacher lesson plan.

What learners will be able to do

- Write functions with several parameters, default values and return values
- Explain the difference between print and return, and fix “None” bugs
- Draw shapes and patterns with the turtle module using loops and functions
- Combine random numbers with functions to simulate dice and run game turns
- Build an interactive menu with input, while loops and if / elif / else
- Store, look up, add, update and loop over data in dictionaries
- Take strings apart with indexes, slicing, methods, in and loops over letters
- Store records in a list of dictionaries and find totals and top values
- Write a simple class with __init__, attributes and methods, and create objects
- Test functions with fixed inputs, even in games that use random numbers
- Design, build, playtest and balance an original turn-based battle game

How every lesson works

Watch	A short animated explainer video with captions and read-aloud narration.
Learn	Learn cards with real Python and a type-along build.
Python Lab	Write and run real Python in the browser. Challenges are checked automatically by running the program; hints unlock one at a time.
Quick checks	Predict-the-output and debugging questions with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	A playable game or tool every module. Starter code for learners and a model solution for the teacher.

Printables	Worksheet for every lesson, project brief for every project and a teacher lesson plan with answer keys.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Functions That Give Answers Back** Write your own functions with several parameters and a return value, and use them to build a cricket score calculator.	parameters and return	Build along: cricket helper functions
2	**Turtle Graphics** Draw shapes and patterns with the turtle module, using loops and functions with parameters, and build a Diwali rangoli generator.	turtle, loops, functions	Build along: shapes with one function
3	**Random Functions and Dice Games** Combine random numbers with functions to simulate dice and build a two-player Ludo dice duel with a function for each turn.	random, simulation, turn functions	Build along: dice lab
4	**Project: Turtle Art Studio** Build an interactive drawing program where the user chooses shapes, colours and sizes, with a separate function for each shape.	functions, input, menus	Project: Turtle Art Studio
5	**Dictionaries: Look It Up** Store information as key and value pairs in a dictionary, look values up, add and update them, loop over them, and build a class quiz bank.	dictionaries, keys, values	Build along: canteen menu
6	**Strings in Depth: Hangman** Use indexes, slicing, string methods, in and loops over letters to take text apart, and build a Hangman game.	slicing, string methods, in	Build along: word explorer
7	**Records and Classes: Virtual Pet** Store records in a list of dictionaries, write your first class with attributes and methods, and build a virtual pet game.	lists of dicts, classes	Build along: cricket team sheet
8	**Capstone: Battle Arena** Design, build and playtest a turn-based text battle game with fighter stats in dictionaries, functions for attack and heal, random damage, a game loop and win or lose endings.	game design, functions, data	Project: Battle Arena

Modules

Module 1 · Functions and graphics

Parameters and return values, turtle graphics, random functions and game turns. Project: Turtle Art Studio.

Big question: How do functions let us build bigger programs out of small, testable pieces?

Module 2 · Data and bigger games
Dictionaries, strings in depth, lists of records and simple classes. Capstone: Battle Arena text RPG.

Big question: How do we organise data so that a program can grow into a real game?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop or Chromebook with a modern browser and a keyboard
- The built-in Kitabash Python Lab (runs in the browser with turtle drawing, no install or account needed)
- Paper and pencil for planning shapes, quiz questions and game designs
- A real or online dice for two warm-up games

Before you start

- Python Game Coders Level 1 (or equivalent): print, variables, input, f-strings, if / elif / else, for and while loops, random, lists and a first simple function with def.

Safety and privacy

Programs never ask for real personal information: use first names, nicknames or made-up names only, and never type phone numbers, addresses, school names or passwords into a program or the chat. Battle games use cartoon fantasy creatures only. Breakout rooms are supervised by the teacher. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Python Game Coders · Level 3 (files, APIs and a first app)**.

Kitabash original curriculum.